

Language Instruction



Pick up the mug
and place it on
the plate.

Multi-view Observations



Front



Left



Right

Current Observations



Front

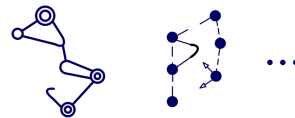


Left



Right

Robot State/Proprioception



PREFIX (visual + language)



MIDDLE (temporal / world)



BOTTLENECK (32 bottleneck tokens)



SUFFIX (state + noise action tokens)



PelicanVLA 0.5 Vision-Language-Action Model

Future-Frame / Future-Latent

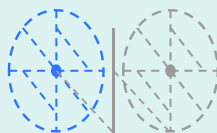


Predict future visual latents



Mechanism Regularizers

Orthogonality Regularization



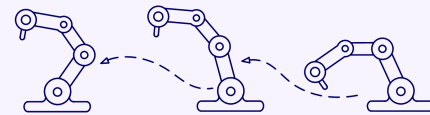
Bottleneck token - Language Alignment



Future-Frame Gating



Action (Trajectory)



Predict action tokens (Δ pose / control)

